

Addendum to “From the Warp/Wormhole Interface to a Throat-Supported Shift Rail”: Geometry Clarification and Current State

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Clarification

The reduced diagnostic model of the May 2026 paper [1] uses the spherically symmetric form

$$ds^2 = -\alpha^2 ds^2 + \gamma_{ll} (dl + \beta^l ds)^2 + \gamma_{\Omega\Omega} d\Omega^2,$$

with the rail coordinate l as its radial direction. Its implementation sets the areal radius by the Ellis profile $R^2 \propto l^2 + R_{th}^2$, a throat of radius $R_{th} = 1.75$ at $l = 0$; a later revision held $R = 1.75$ along the whole track, with flared ends. The geometry is therefore a two-ended, traversable wormhole of Morris–Thorne type. Its lapse stays positive and finite, so light and slow travelers cross the throat directly, and the metric leaves open whether the two flat regions belong to one universe. The earlier system is thus an active rail laid through this throat, and its shortcut combines the throat’s topology with the shift along it. The same shift places the construction relative to the wormhole–warp correspondence [3], where a localized warp bubble crossing a horizon-free throat meets a curvature singularity. The rail’s shift is instead the throat’s own spherically symmetric radial field, written in the proper-distance coordinate with a non-unit lapse, which places it in the regular spherical class of that analysis.

Three readings in the paper and in the accompanying technical disclosure belong to a rail through one space. The disclosure’s service-time comparison with an exterior null signal takes the rail coordinate as the distance between endpoints that lie in separate asymptotic regions, so its ratios compare throat traversal with a reference placed by assumption. Next, the throat relaxation in the service sequence acts on the support stretch, the lapse and an angular capacity jacket, while the minimal sphere persists. Finally, supplied sources hold an existing throat open, whereas forming a two-ended space from ordinary space requires a change of spatial topology [5]. The paper’s two service rules hold unchanged: transport shift stays inside its support envelope, and the packet is caught before release. Its service sequence, prepare $\rightarrow$ carry $\rightarrow$ catch $\rightarrow$ release $\rightarrow$ relax $\rightarrow$ reset, now runs as a literal schedule.

Current state

The current design realizes the rail as a track in one flat space [2], with no throat and unit stretch:

$$ds^2 = -\alpha^2 d\sigma^2 + (dz + \beta d\sigma)^2 + dr^2 + r^2 d\phi^2, \quad \rho = -\frac{1}{32\pi} \left(\frac{\partial_r \beta}{\alpha} \right)^2,$$

with the energy density fixed by the Hamiltonian constraint on the flat slices. The service region $r \leq 1.75$ is the packet-safe region: the shift carries the packet along a prescribed path with a $2.1c$ lane, inside a convex lapse plateau of e^4 whose windows open with the lapse leading and lagging the shift. A transverse boundary layer returns the fields to Minkowski values by $r = 13.25$. Its lapse sheath of e^1 travels with the packet on a schedule, so flat space remains behind each transit, and sources placed along the lane switch on as the packet arrives.

- **Arrival.** The packet reaches the track end at $\sigma = 3.34$. Light through flat space from the same entry event arrives at $\sigma = 5.0$, a mean packet speed of $1.35c$ set entirely by the shift and lapse.
- **Packet.** Its worldtube is timelike from approach to arrival. In the lane the packet moves at $0.016c$ or less relative to local normal observers, and its clock runs at most 55 times exterior time.
- **Admissibility.** Over $\sigma \in [-8, 10]$ and $|z| \leq 10$, every non-vacuum boundary-layer point is Hawking–Ellis Type I, including under refinement, with no Type IV band at any resolved crossing [4]. The exterior is exactly vacuum.
- **Demand.** During a transit the demand is a tension without energy on the shoulders of the lapse profile, local to the packet: peak stress 2.9 and peak negative energy 1.9×10^{-4} , in units where the service region has radius 1.75. At one unit per metre, the peak null-energy deficit integrates to about $0.1 M_\odot$ and the peak stress is 3.5×10^{44} Pa, both independent of route length.

The next stage evaluates candidate source families against this demand.

References

- [1] D. S. Goldman, “From the Warp/Wormhole Interface to a Throat-Supported Shift Rail: A Packet-Centered Active-Rail Architecture after the Wormhole–Warp-Drive Correspondence,” preprint (2026), doi:10.13140/RG.2.2.10402.39368.
- [2] D. S. Goldman, active-rail reports, data and code (2026), including *Geometry Clarification* and *Lapse and Staging Pass*, <https://github.com/dgoldman0/spacetime-metric-engineering>.
- [3] R. Garattini and K. Zatrımaylov, “On the wormhole–warp drive correspondence,” *JCAP* 08 (2024) 061.
- [4] A. T. Le, “On the boundary cost of source-consistent warp shells,” arXiv:2605.25417 (2026).
- [5] R. P. Geroch, “Topology in general relativity,” *J. Math. Phys.* **8**, 782 (1967).